Who pays income tax in France, and where?

Data storytelling dashboard built with Streamlit — Individual project, Data Analysis & Visualization (Prof. Mano Joseph Mathew), EFREI Paris, Data Engineering (DE2), 2026-2027.

1. Context and questions

In 2024, only 3.4% of French tax households paid 53% of all income tax, and fewer than half of households (47%) paid any income tax at all. This report explains how I got to these numbers and what they mean.

I will soon do an internship at the DGFiP (the French public finance administration) in a tax collection office. This office collects the tax that has been issued and follows up with people who do not pay. Before starting, I wanted to understand income tax in numbers: who pays it, and where.

Audience: staff of a tax collection office, or students discovering the topic.

Questions the app answers:

1. How have income tax and the number of taxed households changed since 1984?
2. Which departments pay the most, and where are there the fewest taxed households?
3. How much of the tax relies on high incomes?
4. For a collection office: how many files must be handled to recover most of the money? (simulation)

Story arc: hook (a few households pay most of the tax) → context (the data and its limits) → analysis (trends, departments, income brackets) → insights and implications for tax collection.

2. Data

I use one public dataset: L'impôt sur le revenu par collectivité territoriale (IRCOM), published by the DGFiP on data.gouv.fr under the Open Licence 2.0 (Etalab).

The dataset page has 34 files. I use 8 of them:

File	Format	Years	Content
Impôt sur les revenus de 1984 à 2020 par département	Excel (.xlsx)	1984-2020	One row per department and per year, in euros
IRCOM 2019 to IRCOM 2025 (7 files)	Zip of .xls files, one per department	Income 2018-2024	Detail by income bracket (8 brackets), in thousands of euros

Note: "IRCOM 2025" is published in 2025 and covers 2024 income.

Variables used: number of tax households, reference taxable income (RFR), net income tax, number of taxed households. Region names come from a small department table I added (`data/departements.csv`), and the map uses department outlines from [france-geojson].

Granularity: 101 departments (96 mainland + 5 overseas) plus non-residents, 41 years for the long series, 8 income brackets for 2018-2024.

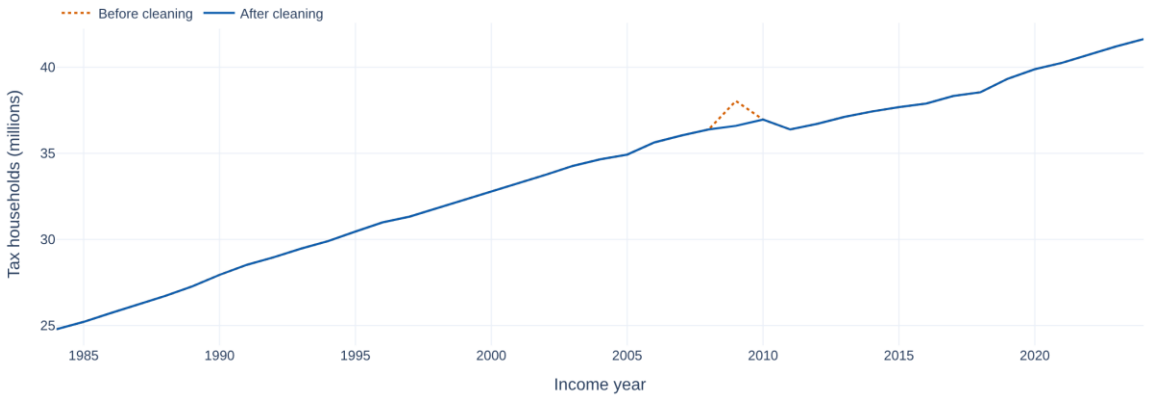
The DGFiP does not publish CSV files. My script `download_data.py` downloads the original files and copies the tables, without changing any value, into two CSV files (3,822 and 6,408 rows). All cleaning is done afterwards in `utils/prep.py`.

3. Data cleaning and quality checks

The raw files had several problems; the biggest one was Paris counted twice in 2009 (1,444,130 extra households).

What I fixed:

- Paris counted twice in 2009: the file has the "Paris" row and also its 5 sub-directions. I removed the 5 duplicate rows.
- Sub-directions before 2010 (Paris-Centre, Nord-Lille, Bouches-du-Rhône-Aix...): 150 rows merged into their department.
- DGFiP codes (`010`, `2A0`, `750`) converted to standard department codes (`01`, `2A`, `75`).
- Units: bracket files are in thousands of euros, converted to euros.
- 3 different file layouts for the bracket zips (2018, 2019-2021, 2022-2024), all read by the same function.
- 2018 ("blank year"): the file has 2 extra columns for the CIMR tax credit. I use the tax before CIMR, so 2018 can be compared with other years.
- Statistical secrecy: 56 cells marked "n.c." are left empty. I did not invent values.
- Ratios (share of taxed households, average tax...) are always computed as sum / sum, never as an average of ratios.



Total number of tax households before cleaning (orange, dotted) and after (blue). The 2009 spike is the Paris double count.

Automatic checks (run at every load and shown in the Conclusions tab):

Check	Result	Detail
Duplicates	OK	0 in the 1984-2020 series, 0 in the 2018-2024 brackets
Double counting department / sub-directions	Fixed	2009, Paris: 5 rows, 1,444,130 households removed
Sub-directions merged	OK	150 rows merged into their department
Statistical secrecy (n.c.)	Flagged	56 hidden cells, left empty
Taxed households ≤ households	OK	0 inconsistent rows
Sum of brackets = Total row	OK	maximum gap: 0.000%
Coverage	OK	100 to 102 territories per year (non-residents from 2006, Mayotte from 2013)
Non-residents in the bracket files	Flagged	missing in 2018 and 2019 (present in the long series)
Series breaks (income ± 15% in one year)	Flagged	2006 (+30%), due to the tax reform

Known series breaks are drawn as dotted lines on the time charts: 2000 (taxable income replaced by RFR), 2006 (tax scale reform), 2018 (CIMR) and 2019 (withholding tax).

4. The app

The app has a sidebar for all controls, a row of 4 KPIs, and 4 tabs that follow the story. It is available in French and English.

Sidebar (top to bottom): language choice, student information, filters, then course information and the EFREI logo.

Filter	What it changes
Period (income years)	Time range of all trend charts
Year for the map and KPIs	Year used by the KPIs, the map, the ranking and the brackets
Scope	Mainland France, overseas, non-residents
Regions	Limits everything to one or more regions
Indicator	Share of taxed households, net tax, average tax, average income, average tax rate, number of households
Department	Detail by bracket and simulation for one department

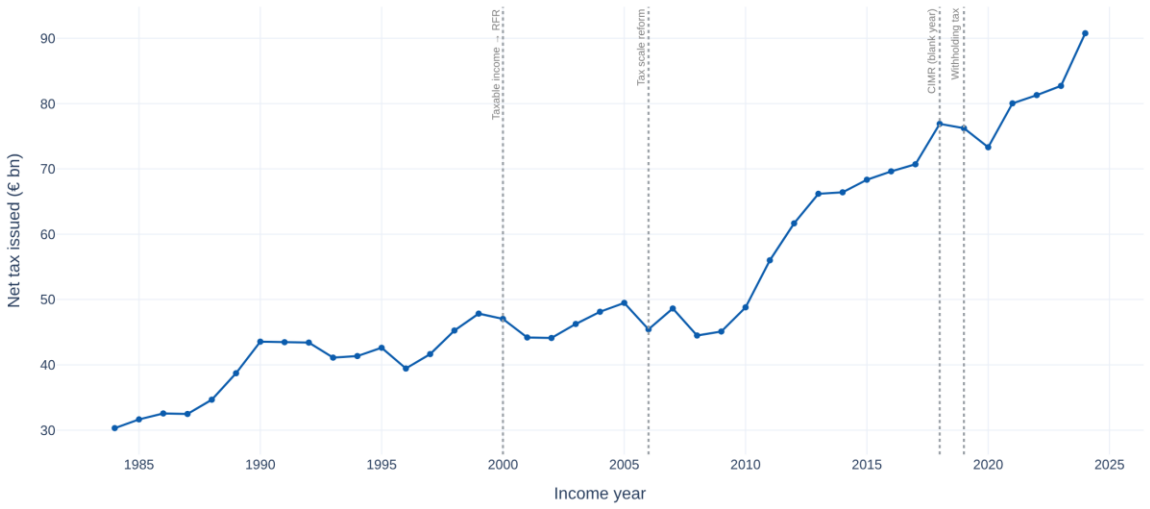
KPIs (linked to the filters): number of tax households, share of taxed households, net tax issued, average tax per taxed household, each with the change versus the year before. For 2024 (mainland + overseas): 41.34 M households, 47.3% taxed, €90.8 bn of net tax, €4,641 per taxed household.

Tab	Question	Main visuals
Introduction	Why this topic? What data?	Text, caveats, glossary
Overview	How has the tax changed since 1984?	Trend line with series breaks, taxed vs not taxed, average tax
Deep dives	Where? Who pays? Which files first?	Map, ranking, scatter plot, brackets, map of high incomes, simulation
Conclusions	Can we trust the numbers? So what?	Checks table, before/after chart, missing values, key insights

Every chart is interactive (Plotly, values on hover), has labelled axes and units, and has a short text description under it.

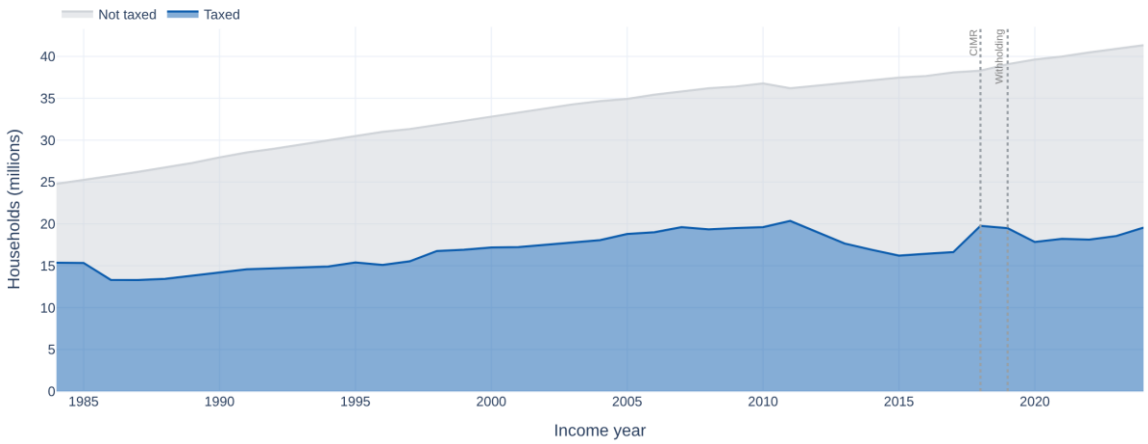
5. Overview: trends since 1984

Between 2000 and 2024, net income tax almost doubled (€47.0 bn → €90.8 bn, +93%) while the number of households grew by only 26% (32.8 M → 41.3 M).



Net tax issued per income year (mainland + overseas). Dotted lines: series breaks. The strong rise starts after 2010.

The share of taxed households goes down over time: 62% in 1984, 52% in 2000, a low point of 43% in 2015, and 47% in 2024. In 2024, 19.6 million households paid income tax out of 41.3 million.



Taxed households (blue) and non-taxed households (grey), in millions. The number of taxed households changes little; most of the growth is in non-taxed households.

So the money comes from a stable group that pays more and more: the average tax per taxed household went from €2,737 in 2000 to €4,641 in 2024 (+70%).



Average net tax per taxed household, in current euros (not adjusted for inflation).

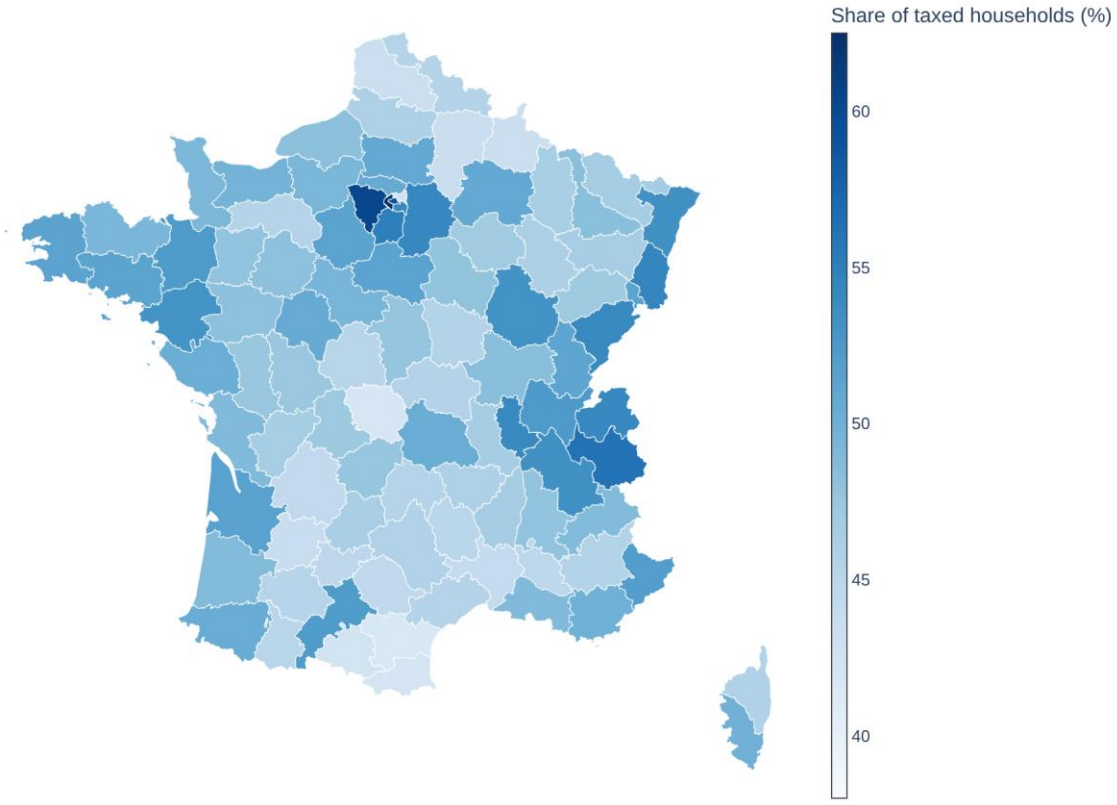
Why so many households pay no income tax: the first part of income is taxed at 0%, the family quotient divides income by the number of shares, a tax reduction (décote) cancels small taxes, tax credits are refunded, and a tax below €61 is not collected.

6. Deep dives (2024)

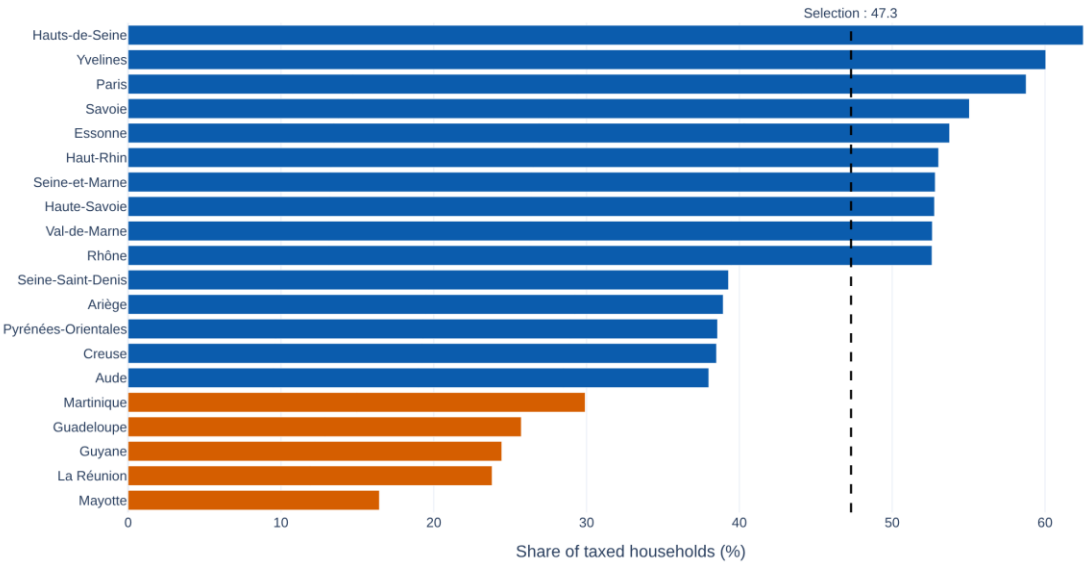
Part 6.1 works for every year from 1984 to 2024. Parts 6.2 and 6.3 need the detail by income bracket, which I loaded only for 2018-2024 income: for an earlier year, the app hides these two parts and shows a message asking to choose a year between 2018 and 2024. All figures below are for 2024.

6.1 Comparing departments

In 2024, the share of taxed households goes from 16.4% (Mayotte) to 62.5% (Hauts-de-Seine); in mainland France the lowest is Aude (38.0%).



Mainland France: the darker the blue, the higher the share of taxed households. The Île-de-France area and the Alps stand out; the north and the south-west Mediterranean coast are lighter.



Highest and lowest departments. Overseas departments in orange. Dashed line: value for the whole selection (47.3%).

The tax is also concentrated in a few places: Paris, Hauts-de-Seine, Yvelines, Rhône and Bouches-du-Rhône pay 33% of the tax, and 13 departments pay half of it.

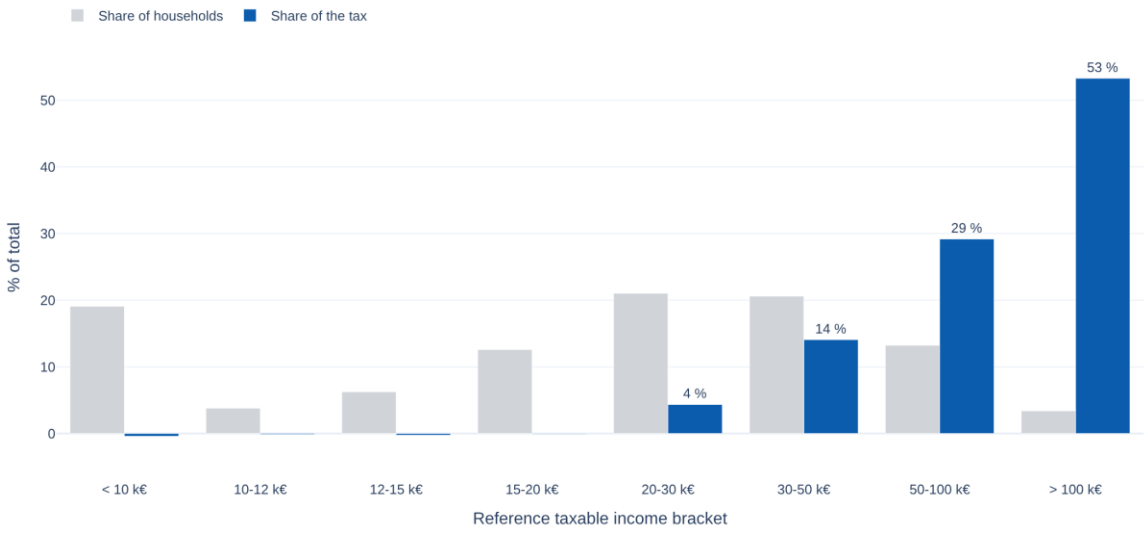


One dot per department, dot size = tax paid, colour = region. Correlation: 0.83.

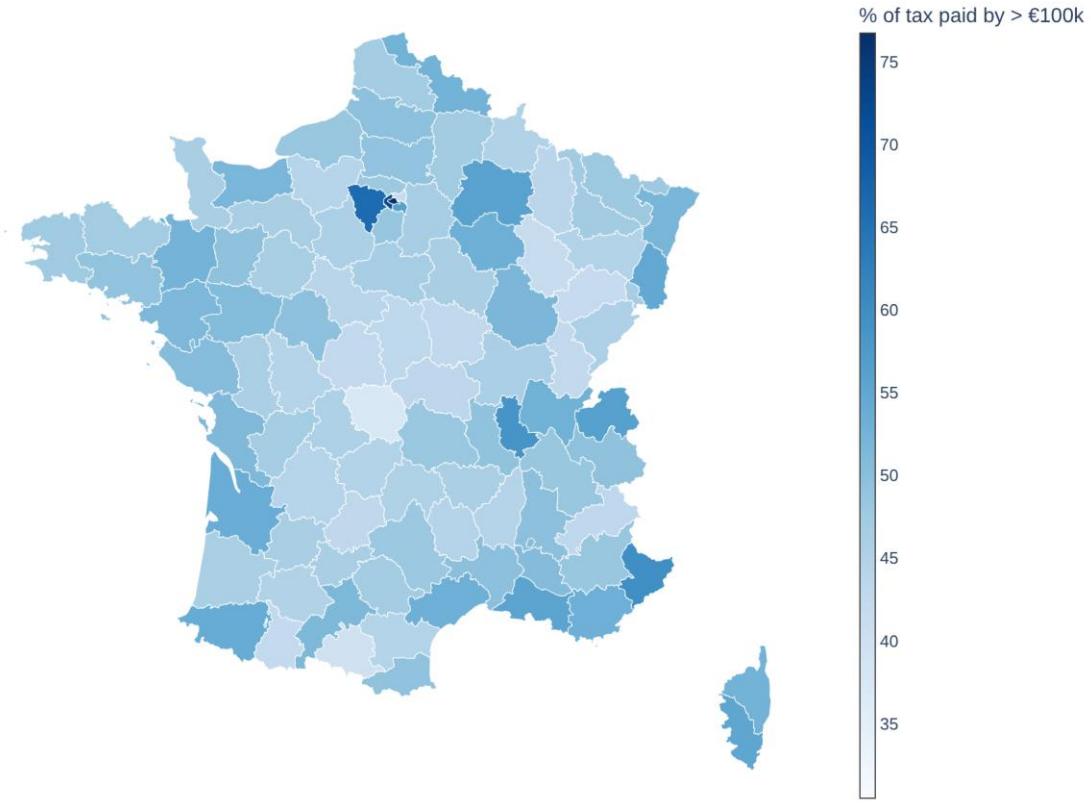
Departments with a higher average income have more taxed households (correlation 0.83). This is expected because income tax is progressive, but a correlation alone does not prove a cause.

6.2 Who pays the tax? Detail by income bracket

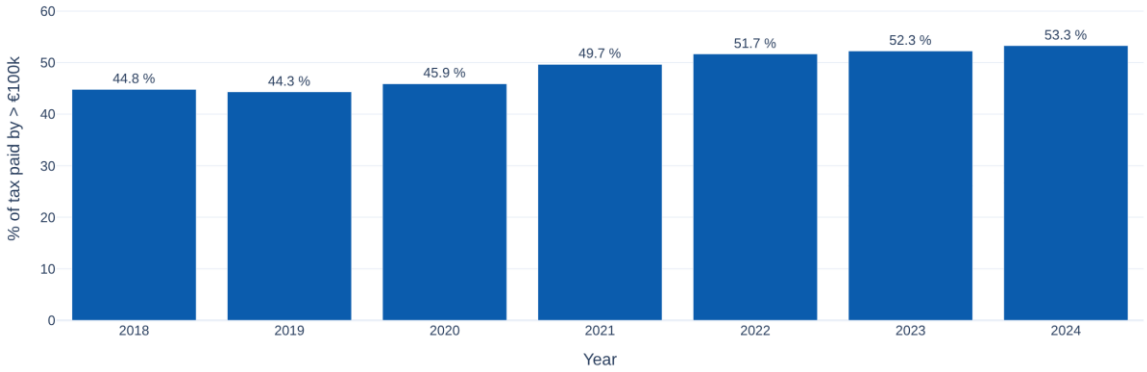
Households above €100,000 of reference income are 3.4% of households but pay 53.3% of the tax.



For each bracket: share of households (grey) and share of the tax (blue). Below €20,000 the net tax is slightly negative because refunded tax credits are larger than the tax.



This share goes from 30% (Creuse) to 77% (Paris).

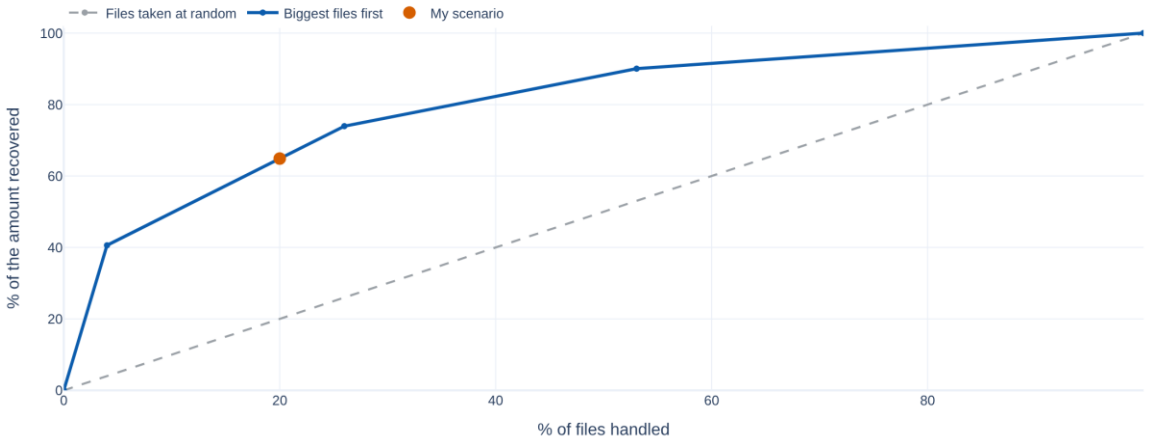


From 44.8% in 2018 to 53.3% in 2024. Brackets are not adjusted for inflation, so part of this rise only comes from incomes going up. Non-residents are missing from the 2018 and 2019 bracket files.

6.3 Simulation: which files should be handled first?

With my assumptions, handling only the biggest 20% of late files recovers about 65% of the unpaid amount.

The data does not say how much tax is unpaid, so the simulation uses unpaid rates that I chose (sliders in the app): 3% for income below €30k, 1.5% for €30-100k, 1% above €100k. With these values, for 2024 (mainland + overseas), there are about 340,000 late files for €1.19 bn.



Blue: files sorted from the biggest amount to the smallest. Grey dashed line: files taken at random. Orange dot: my scenario (20% of files → 65% of the amount).

Method: late files = taxed households × unpaid rate, amount = net tax × unpaid rate, by bracket; brackets with a negative tax are removed. I assume all files in one bracket have the same amount, so real concentration is probably even stronger.

7. Key insights and implications

Income tax is concentrated twice: on a few households and on a few places.

1. Concentrated on high incomes: in 2024, the 3.4% of households above €100,000 pay 53% of the tax, while 53% of households pay no income tax.
2. Concentrated geographically: 13 departments pay half of the tax. Île-de-France has 18% of households but 36% of the tax (all territories, non-residents included, as in the app).
3. Each file weighs more: the number of taxed households is stable, but the average tax per taxed household rose by 70% between 2000 and 2024.

What this means for a tax collection office (my opinion):

- follow the big files closely, because they hold most of the money;
- automate reminders for small amounts, which are very numerous;
- the means needed are not the same in every department.

Next steps:

1. Find data on tax actually collected (collection rates) to replace my assumptions.

2. Go down to municipality level (the IRCOM files allow it).
3. Add local taxes for a complete picture.
4. Adjust amounts for inflation.

8. Limitations and ethics

- The data shows tax **issued**, not tax **collected**: unpaid tax is not visible, so the simulation relies on assumptions.
- Amounts are in current euros, and income brackets are not adjusted for inflation.
- Before 2000 the published income is net taxable income, not RFR, so income levels are not directly comparable.
- 2018 and 2019 are special years (CIMR, then withholding tax).
- Capital income taxed at the flat rate (PFU) is not included.
- Department level only: differences inside a department are not visible.
- Correlation is not causation.
- The data is aggregated by the DGFiP and protected by statistical secrecy: no personal data is used, and there is no risk of re-identification.

9. Technical choices and how to run it

The code follows the structure recommended in the project instructions:

File	Role
app.py	Main page: language, sidebar filters, KPIs, tabs
sections/	intro.py, overview.py, deep_dives.py, conclusions.py
utils/io.py	Download and read the IRCOM files
utils/prep.py	Cleaning, indicators, quality checks
utils/viz.py	Chart functions with one common style
utils/lang.py	t("français", "english") for the French / English switch
download_data.py	Rebuilds the two CSV files from data.gouv.fr

- **Performance:** loading and cleaning are cached with st.cache_data; the app reads two small CSV files (about 1 MB) instead of hundreds of Excel files.
- **Map:** each department is drawn with Plotly from a GeoJSON file, so the map needs no external background map.
- **Accessibility:** colour-blind-friendly colours, labelled axes and units, hover tooltips, and a text description under every chart.
- **Reproducibility:** pinned versions in requirements.txt (Streamlit 1.66, pandas, Plotly); no random sampling, so results are always the same.

To run it locally:

```
pip install -r requirements.txt
streamlit run app.py
```

Links

- App online: ircom-impot-revenu-app.streamlit.app
- Code: github.com/zanzibarrr/ircom-impot-revenu-streamlit
- Dataset: IRCOM on data.gouv.fr (DGFiP, Open Licence 2.0)
- Department outlines: france-geojson
- Demo video: in the ffolder